

12. Real and Ideal Computers

1. Symbols and numbers.
2. Error and accuracy.
3. Finiteness and infinity.
4. Actuality and possibility.
5. Competence and performance.
6. Computing a function vs. instantiating (or satisfying) a function.

“Humans, in particular, routinely satisfy functions by computing them.”

(Cummins 1991, p. 91)

“But + is a function whose arguments and values are numbers; and whatever numbers are, they are not states or processes or events in any physical system. How, then, can a physical system be described by +? How can a physical system traffic in numbers and hence add? The answer, of course, is that numerals—i.e., representations of numbers—can be states of a physical system, even if the numbers themselves cannot. A physical system adds by trafficking in numerals and hence, indirectly, in the numbers those numerals represent.” (Cummins 1991, p. 89)

“From an explanatory point of view, what interpretation provides is a link between mere state crunching (button-pressing-to-display transitions) and addition: Under interpretation, the state transitions of the system are revealed as adding. You can get a feel for the explanatory role of interpretation by imagining that the device is an archaeological discovery. Perhaps you know from documents that the thing is an adding machine; however, looking at it, you have no idea, at first, how to give inputs or read outputs. You can “make it go,” let’s say, by turning a handle, but this is just more or less random manipulation until you find an interpretation that reveals the state transitions of the device as sum calculations and hence establishes the representational status and contents of those states. To know how to use it is to know how to see it as simulating +, and that means knowing how to “make it go” and how to interpret it.” (Cummins 1991, p. 94)